

kmaps

P2

ab\cd	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	1	0	1	0
10	1	0	1	0

msop

$$f(a,b,c,d) = a\bar{c}\bar{d} + \bar{a}\bar{c}d + acd + \bar{a}c\bar{d}$$

$$= a(\bar{c}\bar{d} + cd) + \bar{a}(\bar{c}d + c\bar{d})$$

$$\rightarrow A \text{ xor } (C \text{ xor } D)$$

Sum
long

P1

ab\cd	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

msop

$$f(a,b,c,d) = \bar{a}b\bar{c} + a\bar{b}\bar{c} + \bar{a}b\bar{c} + abc$$

$$= a(\bar{b}\bar{c} + bc) + \bar{a}(b\bar{c} + \bar{b}c)$$

$$\rightarrow A \text{ xor } (B \text{ xor } C)$$

P0

ab\cd	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	1	0	1	0
10	0	1	0	1

msop

$$f(a,b,c,d) = b\bar{c}\bar{d} + \bar{b}\bar{c}d + bcd + \bar{b}c\bar{d}$$

$$= b(\bar{c}\bar{d} + cd) + \bar{b}(\bar{c}d + c\bar{d})$$

$$\rightarrow B \text{ xor } (C \text{ xor } D)$$

$$P2 \rightarrow A \text{ xor } (C \text{ xor } D)$$

$$P1 \rightarrow A \text{ xor } (B \text{ xor } C)$$

$$P0 \rightarrow B \text{ xor } (C \text{ xor } D)$$

xor representation

ModelSim - INTEL FPGA STARTER EDITION 2020.1

EditViewCompileSimulateAddWaveToolsLayoutBookmarksWindowHelp

100 ps

LayoutSimulate

Search:

columnLayoutAllColumns

sim - Default

Instance	Design unit	Design unit type
testbench	testbench(b...	Architecture
hamming_0	hamming(rtl)	Architecture
testbench_proc...	testbench(b...	Process
standard	standard	Package
textio	textio	Package
std_logic_1164	std_logic_1...	Package
numeric_std	numeric_std	Package

Processes (Active)

Name	Type (filtered)
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Objects

Name	Value	Type
d_signal	1110	Signal
p_signal	010	Signal
p_expected	010	Signal

second to last iteration
4 data bits mapping to 010 parity

Wave - Default

/testbench/d_signal

1110

/testbench/p_signal

010

/testbench/p_expec...

010

d signal -> data bits

p_signal -> xor set output

p_expected -> manual output using data bit to parity map

Msgs	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
	000	101	111	010	011	110	100	001	110	011	001	100	101	000	010	111
	000	101	111	010	011	110	100	001	110	011	001	100	101	000	010	111

from data bit value 0 to 15; resulting outputs match mapping of parity bits
timing offset by 10 ns per test vector, no indication of errors

Now

180 ns

Cursor 1

146.009 ns

Transcript

- Compiling architecture behavioral of testbench

End time: 11:42:11 on Oct 06, 2023, Elapsed time: 0:00:00

Errors: 0, Warnings: 0

sim work.testbench

Start time: 11:42:11 on Oct 06, 2023

loading std.standard

loading std.textio(body)

loading ieee.std_logic_1164(body)

loading ieee.numeric_std(body)

loading work.testbench(behavioral)

loading work.hamming(rtl)

* Note: Number of errors = 0

Time: 160 ns Iteration: 0 Instance: /testbench

0 ps to 190 ns

0 ps to 190 ns

total runtime 160 ns with 0 errors across test vectors

15 test cases for all combinations of 4 data bit inputs

2:52 AM

10/7/2023